Appendix 1. Alternative approach to addressing missing outcome data

To decide whether to rate down certainty of evidence for risk of bias in randomized control trials, Core GRADE suggests systematic reviewers use one of the available instruments to assess risk of bias associated with each item for each included study. Ultimately, these ratings inform judgments of the overall risk of bias of each individual study as low or high.

For missing outcome data, one instrument recommended for Core GRADE, ROBUST-RCT,[1] identifies two possible approaches. One is simple: set thresholds for percent missing data for each rating of risk of bias (e.g. missing data < 5% low risk of bias, 5% to 10% probably low, 10% to 15% probably high, over 15% high risk of bias). A second more sophisticated approach involves systematic reviewers conducting a primary meta-analysis using complete-case analysis (i.e., include only the participants whose outcome data are available) followed by sensitivity meta-analyses imputing, for each individual study, data for the participants whose outcome data are unavailable.

The aim of the approach is to test the robustness of the inference made by the primary meta-analysis. For example, for an analysis of a binary outcome, if a primary meta-analysis suggests a beneficial effect of an intervention (i.e., the intervention decreases risk of an undesirable outcome), reviewers may assume the event rates in the comparator group in those followed successfully and in those with missing data are the same. In contrast, the sensitivity analyses will challenge the robustness of the inference of beneficial intervention effect by assuming in the intervention group the event rate in the participants with missing data is higher than the event rate in those whose outcome data are available. The ratio of the two event rates may range from 1.5 (the least challenging assumption) to 5 (the most challenging assumption).[2] Systematic reviewers need to decide which strategy represents the most extreme but still plausible in their specific context.

For continuous outcomes, reviewers need to impute both means and standard deviations for participants with missing data in each group. If a primary meta-analysis suggests a beneficial effect of intervention, the serial imputations attribute progressively worse outcomes to those in the intervention group with missing data, and progressively better outcomes in comparator group with missing data. The sensitivity analyses thus provide progressively more stringent criteria for inferring robustness of the results.[2] Once again, systematic reviewers need to decide which strategy represents the most extreme but still plausible in their specific context. Systematic reviewers can adopt similar approach to addressing risk of bias related to missing data for time-to-event outcomes.[3]

The ultimate goal of this approach is to decide whether to rate down certainty of body of evidence for missing outcome data. If the inference is robust to the most extreme assumption viewed as plausible, one does not need to rate down. Otherwise, rating down is warranted.

This approach addresses risk of bias associated with missing data across the entire body of evidence at the meta-analysis level rather than at individual study level. As described above, it is more sophisticated and not so familiar to systematic reviewers. Thus, we consider this approach as an alternative beyond Core GRADE.

References

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Appendix 2. Alternative approach to rating up certainty of evidence from non-randomized studies of interventions

Up to now, guidance from the GRADE Working Group has taken one of two approaches to assessing certainty of evidence from non-randomized studies of interventions (NRSI):

i) Given the seriousness of the residual confounding problem, all NRSI start as low certainty evidence but can be rated down to very low or be rated up to moderate or high. This approach is valid, familiar to most GRADE users, and therefore we opted for recommending it as the primary approach in Core GRADE.

ii) Using ROBINS-I,[1] starting all NRSI as high certainty evidence.[2] The complexity of ROBINS–I is inconsistent with the objectives of Core GRADE.

However, in discussing this topic, we found ourselves also acknowledging that the primary GRADE position of always considering NRSI to start as low certainty has been one of the aspects of GRADE creating barriers to its adoption, particularly in fields where RCTs are more difficult to perform and therefore RCT based evidence less common, including public health, nutrition, toxicology, and rare diseases. This was one of the motivations for the creation of ROBINS–I and the enthusiasm of some GRADE working group members for its use.

Therefore, we have conceived and will present here an alternative conceptual framework to the process of rating up certainty of evidence from NRSI. This approach might appeal particularly to systematic reviewers, guideline developers and health technology assessors mostly relying on NRSI, including routine clinical observation.

The key difference between the primary approach presented in the main text and the alternative approach presented here is that the alternative one uses considerations about large effect and dose-response gradient not to rate up the certainty, but to establish the starting point for the certainty of evidence from NRSI. That is, RCT will always start at high certainty, and NRSI will start low unless they present a large effect or a dose-response gradient. Specifically, NRSI presenting a large effect (RR >2.0 or <0.5) or a gradient will start at moderate certainty. NRSI presenting a very large effect (RR > 5.0 or <0.2) or presenting any large effect and a dose-response gradient at the same time will start at high certainty. Thereafter, one will determine whether rating down certainty is needed or not by assessing risk of bias with criteria appropriate for RCTs or NRSI.

There are potential upsides to this alternative approach. First, it follows a more linear reasoning. Considering NRSI starting low in all cases, only to later modifying our judgment and recognize some are actually providing higher certainty may be perceived as a convoluted and contradictory process. One might consider it more logical and appropriate to set the starting level certainty looking at the characteristics of the study, beyond the simple aspect of randomization. This approach does not, as in ROBINS–I, suggest allowing liberally setting the certainty. However, it does use the large effect and dose-response gradient concepts not to “rate up” after rating down, but to set an appropriate starting point, in a somewhat more respectful and fair consideration of the quality and value of those specific studies and their authors. It is in the nature of the effect to be large or to present a gradient, and that strengthens the corresponding evidence, which therefore obviates the need to set a low certainty starting point.

Second, it takes away an unresolved and somewhat embarrassing situation which happens with the rating up mechanism, which is the decision to rate up (or not) after having rated down the same evidence during the RoB assessment. On this point, GRADE is not proposing a unique solution, leading to the same body of evidence potentially being assigned a different level of certainty depending on one rating up or not after having rated down. By using the large effect and gradient criteria to set the starting certainty point, the issue is naturally solved, as rating down will happen at a later step, with no ambiguity.

Third, the two criteria were empirically discovered. They are not common but present from time to time. Setting the starting level of certainty basing the judgement on empirically established criteria is a logical approach, which is very similar in nature to the concept of target trial,[3] or the assumption behind the logical construct of ROBINS-I.
We anticipate that those systematic reviewers, guidelines users and health technology assessors sympathetic with ROBINS-I but hesitating because of its complexity will find the alternative approach more natural and congenial. We are not suggesting, at this time, that in keeping with ROBINS–I, NRSI will start high or moderate for reasons different from large effects or gradient. However, should empirical evidence be put forward to make the case that other study characteristics might contribute to increase our certainty, they could be incorporated in future guidance.

Fourth, it is important to outline that the proposed alternative approach will lead exactly to the same results as the rating up process, thus not causing any disagreement or deviation in the appraisal of the same body of evidence.

The major downside we see to this approach, which we considered important enough to propose it as an alternative and not as the main approach, is the risk that it would appear as a significant deviation from current GRADE practice, and would therefore be unwelcome to long standing GRADE users. Also, we do feel that, while many may perceive the alternative approach as promising, organic, and logical on the theoretical plane, it has not been piloted and has not survived yet the empirical test of repeated application.

For all those who will opt to adopt the proposed alternative approach, we suggest they do mention it explicitly in their publications, Summary of Findings tables, and reports. Reporting on upsides and downsides experienced in applying the approach will help establishing the value of this alternative approach.

References

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